

Bluetooth Mesh: Putting the Smartness in Smart Buildings

This article is a part of the speech Martin Woolley delivered at India Electronics Week, February 2019, held in Bengaluru. Using his wit and humour, he explains the intricacies of the latest Bluetooth technology and its applicability for lighting and other purposes.



Bluetooth is a standard. This means you can buy products from different manufacturers and these will work together because of the standardisation process. Bluetooth has been around for quite some time now. The key to Bluetooth's lasting success is that it has not stood still. It has been systematically developed over the years to keep pace with market requirements. Its usage has become broader and broader, and it has found its way into different types of devices and industry sectors. All these have enabled Bluetooth to survive till now.

About ten million Bluetooth devices ship every day, and the number

is growing. This is quite remarkable. In the summer of 2017, the group released the specification for Bluetooth Mesh, version 1.0. It is a completely new way of using Bluetooth for smart buildings.

Bluetooth technology has evolved in terms of topologies. In the beginning, it was a cable-replacement technology. It was about taking serial cables between two devices, throwing those away and replacing them with wireless communication. Now, we are talking about many-to-many communication between thousands of devices. That is what mesh networking is all about. And it will be used in all types of industry

sectors. We probably do not even know what those sectors are because the marketplace always surprises us in the way it uses technology.

Types of Bluetooth radios

There are two distinct Bluetooth radios. The first is called Bluetooth Basic Rate/Enhanced Data Rate (BR/EDR), which is the Bluetooth. It is best used for steady streams of data and very successful in the world of wireless audio. It was released twenty years ago, and is still going strong.

The second is Bluetooth Low Energy, which was released in 2010. It has a completely different design with different